

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied business analytics

Practical – III Task

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Code:

```
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

# Load Dataset
df = pd.read_csv("house_price_regression_dataset.csv")

# -----
# Q.1 Correlation between Year_Built and House_Price
# -----
corr_q1 = df["Year_Built"].corr(df["House_Price"])
print("Q.1 Correlation (Year_Built vs House_Price):", round(corr_q1, 4))

# -----
# Q.2 Correlation Matrix for all Numerical Columns
# -----
corr_matrix = df.corr(numeric_only=True)
print("\nQ.2 Correlation Matrix:")
print(corr_matrix.round(4))

# -----
# Q.3 & Q.4 Model 1: Year_Built -> House_Price
# -----
X1 = df[["Year_Built"]]
y = df["House_Price"]

model1 = LinearRegression().fit(X1, y)
pred1 = model1.predict(X1)
r2_model1 = r2_score(y, pred1)

print("\nQ.3 Model 1 (Year_Built -> House_Price):")
print(" Coefficient (Slope):", round(model1.coef_[0], 4))
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print(" Intercept:", round(model1.intercept_, 4))
print("Q.4 R2 Score (Year_Built -> House_Price):", round(r2_model1, 4))

# -----
# Q.5 Model 2: Square_Footage -> House_Price
# -----
X2 = df[["Square_Footage"]]

model2 = LinearRegression().fit(X2, y)
pred2 = model2.predict(X2)
r2_model2 = r2_score(y, pred2)

print("\nQ.5 R2 Score (Square_Footage -> House_Price):", round(r2_model2, 4))

# -----
# Q.6 Model 3: Neighborhood_Quality -> House_Price
# -----
X3 = df[["Neighborhood_Quality"]]

model3 = LinearRegression().fit(X3, y)
pred3 = model3.predict(X3)
r2_model3 = r2_score(y, pred3)

print("\nQ.6 R2 Score (Neighborhood_Quality -> House_Price):", round(r2_model3, 4))

# -----
# Q.7 Comparison of R2 Scores
# -----
print("\n=== Q.7 Comparison of R2 Scores ===")
print(f"1. Square_Footage -> House_Price:      R2 = {r2_model2:.4f} ({r2_model2*100:.2f}%)")
print(f"2. Year_Built -> House_Price:          R2 = {r2_model1:.4f} ({r2_model1*100:.2f}%)")
print(f"3. Neighborhood_Quality -> House_Price: R2 = {r2_model3:.4f} ({r2_model3*100:.2f}%)")

```

Output:

Q.1 Correlation (Year_Built vs House_Price): 0.052

Q.2 Correlation Matrix:

	Square_Footage	Num_Bedrooms	Num_Bathrooms	Year_Built
Square_Footage	1.0000	-0.0436	-0.0316	-0.0224
Num_Bedrooms	-0.0436	1.0000	0.0228	-0.0158
Num_Bathrooms	-0.0316	0.0228	1.0000	-0.0211
Year_Built	-0.0224	-0.0158	-0.0211	1.0000
Lot_Size	0.0895	-0.0094	0.0349	-0.0611
Garage_Size	0.0306	0.1138	0.0248	-0.0255
Neighborhood_Quality	-0.0084	-0.0490	0.0176	-0.0095
House_Price	0.9913	0.0146	-0.0019	0.0520

	Lot_Size	Garage_Size	Neighborhood_Quality	House_Price
Square_Footage	0.0895	0.0306	-0.0084	0.9913
Num_Bedrooms	-0.0094	0.1138	-0.0490	0.0146
Num_Bathrooms	0.0349	0.0248	0.0176	-0.0019
Year_Built	-0.0611	-0.0255	-0.0095	0.0520
Lot_Size	1.0000	0.0024	0.0376	0.1604
Garage_Size	0.0024	1.0000	-0.0113	0.0521
Neighborhood_Quality	0.0376	-0.0113	1.0000	-0.0078
House_Price	0.1604	0.0521	-0.0078	1.0000

Q.3 Model 1 (Year_Built -> House_Price):

Coefficient (Slope): 638.6525

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=== Q.7 Comparison of R2 Scores ===

1. Square_Footage -> House_Price: R2 = 0.9826 (98.26%)
2. Year_Built -> House_Price: R2 = 0.0027 (0.27%)
3. Neighborhood_Quality -> House_Price: R2 = 0.0001 (0.01%)